Geometry Right Triangles & Trigonometry Cheat Sheet

More Practice: SOH-CAH-TOA and inverse trig

1 TRIG RATIOS

In a right triangle, the trigonometric ratios relate an angle to the lengths of the sides.

$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan\theta = \frac{\text{opposite}}{\text{adjacent}}$$

Memory Aid:

SOH - CAH - TOA

Sine $= \frac{\text{Opposite}}{\text{Hypotenuse}}$

Cosine $= \frac{\text{Adjacent}}{\text{Hypotenuse}}$

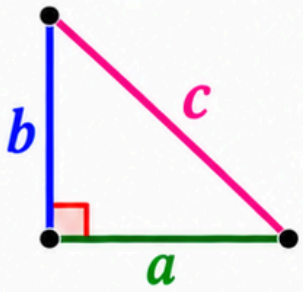
Tangent $= \frac{\text{Opposite}}{\text{Adjacent}}$

2 PYTHAGOREAN THEOREM

In a right triangle, the squares of the lengths of the legs a and b add up to the square of the hypotenuse c.

$$a^2 + b^2 = c^2$$

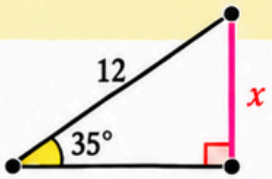
- a and b are the **legs** (the two sides that form the right angle).
- c is the **hypotenuse** (the side opposite the right angle).



3 FINDING A SIDE WITH SINE

Use sine when you know the **opposite** side and the hypotenuse (or need to find the **opposite** side).

Example: Find the length of x.



Step 1: Write the trig ratio.

$$\sin 35° = \frac{x}{12}$$

Step 2: Solve for x.

$$x = 12 \sin 35°$$

Step 3: Use a calculator.

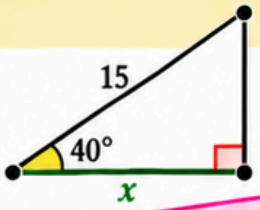
$$x \approx 6.9$$

Answer: $x \approx 6.9$

4 FINDING A SIDE WITH COSINE

Use cosine when you know the **adjacent side** and the hypotenuse (or need to find the **adjacent** side).

Example: Find the length of x.



Step 1: Write the trig ratio.

$$\cos 40° = \frac{x}{15}$$

Step 2: Solve for x.

$$x = 15 \cos 40°$$

Step 3: Use a calculator.

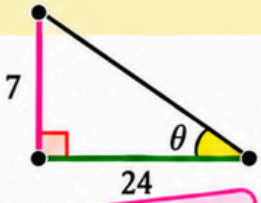
$$x \approx 11.5$$

Answer: $x \approx 11.5$

5 FINDING AN ANGLE WITH INVERSE TANGENT

Use tangent when you know the **opposite** and **adjacent sides** and need to find the angle.

Example: Find the measure of θ.



Step 1: Write the trig ratio.

$$\tan\theta = \frac{7}{24}$$

Step 2: Solve for θ using inverse tangent.

$$\theta = \tan^{-1}\left(\frac{7}{24}\right)$$

Step 3: Use a calculator.

$$\theta \approx 16.3°$$

Answer: $\theta \approx 16.3°$

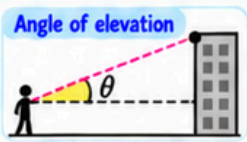
6 CHOOSING THE CORRECT RATIO

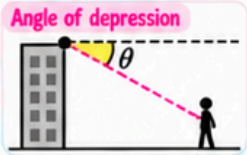
Pick the trig ratio that matches the sides you know (or the side you need to find).

Known Sides	Use This Ratio	
Opposite and hypotenuse	**Sine**	$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}$
Adjacent and hypotenuse	**Cosine**	$\cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}$
Opposite and adjacent	**Tangent**	$\tan\theta = \frac{\text{opposite}}{\text{adjacent}}$

7 ANGLE OF ELEVATION / DEPRESSION

- **Angle of elevation:** the angle formed by a line of sight that looks upward from a horizontal line.
- **Angle of depression:** the angle formed by a line of sight that looks downward from a horizontal line.





The trigonometric work is the same for both. Draw and label the diagram correctly, identify the angle and the sides, then use the appropriate trig ratio (SOH-CAH-TOA).

8 QUICK STRATEGY TIPS & COMMON MISTAKES

QUICK TIPS:

- Set your calculator to degree mode (DEG).
- Draw and label the right triangle.
- Identify the sides (opposite, adjacent, hypotenuse) relative to the chosen angle before selecting a trig ratio.
- Round your final answer as directed (e.g., nearest tenth).
- Include units when appropriate (e.g., cm, m, ft, degrees).

COMMON MISTAKES:

- Using the wrong trig ratio for the given sides.
- Mixing up opposite and adjacent sides.
- Not setting the calculator to degree mode.
- Rounding too early in multi-step problems.
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Memory Aid:

SOH - CAH - TOA

Sine = Opposite / Hypotenuse

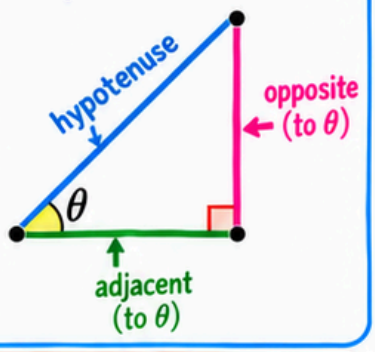
Cosine = Adjacent / Hypotenuse

Tangent = Opposite / Adjacent

2 LABELING SIDES

The names of the sides (opposite, adjacent, hypotenuse) depend on which angle you are using.

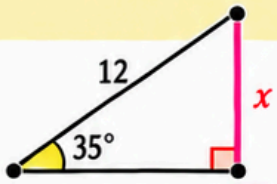
- **Opposite side:** across from the chosen angle (θ).
- **Adjacent side:** next to the chosen angle, but not the hypotenuse.
- **Hypotenuse:** across from the right angle. It is always the longest side.



3 FINDING A SIDE WITH SINE

Use sine when you know the **opposite** side and the hypotenuse (or need to find the **opposite** side).

Example: Find the length of x.



Step 1: Write the trig ratio.

$$\sin 35° = \frac{x}{12}$$

Step 2: Solve for x.

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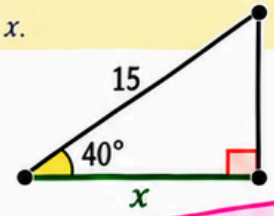
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Answer: $x \approx 6.9$

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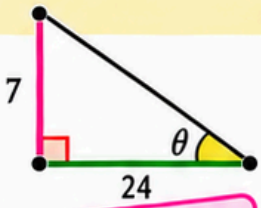
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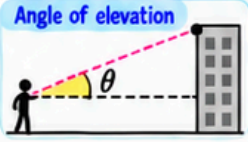
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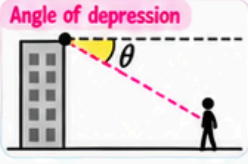
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